Project Part 1:

Small Data Problem Analysis Report

Complete this document and submit it with your project.

Match the scenario with the most appropriate solution and explain your choice

Scenario #1: Travel Planner Problem

A travel planning company asks customers to share pictures of past vacations/holidays so their staff can identify what kind of trips they enjoy. The company offers three basic categories of trips:

- Exploring in the Forest
- Adventure in the Desert
- Relaxing on the Beach

As part of a new online trip planning software, the company is creating an AI bot that will automatically figure out from the uploaded photos which category is likely to be most appealing to the customer. The challenge is the company has fewer than 500 photos that are categorized, and they feel it will be difficult to train a model using such little data.

Scenario #1: Travel Planner Problem Should you use transfer learning or a synthetic data approach to solve this problem? Please explain your answer in a short paragraph containing 3-5 sentences.	**Transfer Learning** **Explanation:** Transfer learning is the best solution for this problem because the dataset contains fewer than 500 categorized images, which is too small to train a deep learning model from scratch effectively. Since this is an image classification problem, a pre-trained model such as VGG16 can be used because it has already learned important image features from large datasets like ImageNet. By freezing most of the existing layers and retraining only the final classification layer, the model

	can achieve high accuracy even with limited data. This approach also reduces training time and helps prevent overfitting.

Scenario #2 Loan Funding Prediction Problem

A loan company has a fairly large dataset that they want to use to train a model that predicts whether or not a loan should be funded. The problem they face is the dataset they are using has a large class imbalance... they don't have enough examples of loans that were denied. This is creating a model that doesn't perform well, particularly for loans that probably should be denied.

Scenario #2: Loan Funding Prediction Problem Should you use transfer learning or a synthetic data approach to solve this problem? Please explain your answer in a short paragraph containing 3-5 sentences.	Synthetic data is the best solution for this problem because the loan dataset has a severe class imbalance, with very few denied loan examples. This imbalance causes the prediction model to perform poorly for the denied loan category. Using a Variational Autoencoder (VAE), additional synthetic denied-loan records can be generated to balance the dataset. After augmenting the original dataset with the generated data, the model can learn patterns more effectively and improve precision, recall, and F1-score for denied loan predictions.